DAYANANDA SAGAR UNIVERSITY, BENGALURU

School of Engineering • End Semester Examination

Course: Analog and Digital Electronics | Course Code: 21CS33 | Credits: 3

Exam Session: End Semester Exam 2024 **Max Marks: 80 | Duration: 3 Hours**

INSTRUCTIONS TO CANDIDATES:

1. Answer FIVE full questions, choosing ONE full question from each module.
2. Draw neat diagrams wherever necessary. Missing data may be suitably assumed.

MODULE 1

Q1. Simplify the Boolean function F(A, B, C, D) = sum m(0, 2, 5, 7, 8, 10, 14, 15) + sum d(3, 11) using a 4-variable Karnaugh Map and realize the simplified expression using basic logic gates. **[8 Marks]**

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Q2. Design a 3-bit Synchronous Up-Counter using JK Flip-Flops. Draw the state diagram, excitation table, K-maps for J & K inputs, and final logic schematic. **[8 Marks]**